

Market Liquidity Measurement Framework

A Comprehensive Model for Trading Book Analysis

Technical Documentation

December 22, 2025

Abstract

This document provides a comprehensive technical description of a market liquidity measurement framework for analyzing trading positions. The framework incorporates multiple established liquidity metrics including bid-ask spreads, the Amihud illiquidity ratio, volume-based measures, and market impact estimation. We present the mathematical formulations, implementation algorithms, and practical applications of this multi-dimensional approach to liquidity analysis.

Contents

1	Introduction	2
2	Mathematical Framework	2
2.1	Position Representation	2
2.2	Bid-Ask Spread Measures	2
2.2.1	Absolute Spread	2
2.2.2	Relative Spread	2
2.2.3	Spread in Basis Points	3
2.3	Amihud Illiquidity Ratio	3
2.4	Volume Liquidity Ratio	3
2.4.1	Time to Liquidate	3
2.5	Market Impact Model	3
2.5.1	Temporary Impact	3
2.5.2	Permanent Impact	4
2.5.3	Total Market Impact	4
2.6	Composite Liquidity Score	4
2.6.1	Spread-Based Score	4
2.6.2	Volume-Based Score	4
2.6.3	Liquidity Rating Classification	4
3	Portfolio-Level Aggregation	4
3.1	Total Portfolio Value	5
3.2	Weighted Average Spread	5
3.3	Total Liquidation Cost	5
3.4	Portfolio Liquidity Cost Ratio	5
3.5	Average Liquidity Score	5

4	Algorithm and Flow Charts	6
4.1	High-Level Analysis Flow	6
4.2	Position-Level Metric Calculation	7
4.3	Market Impact Calculation Flow	7
4.4	Composite Score Computation	8
5	Implementation Algorithm	8
6	Practical Example	8
6.1	AAPL Analysis	8
6.2	MICRO Analysis (Illiquid Position)	10
7	Interpretation and Usage	10
7.1	Position-Level Decisions	10
7.2	Portfolio-Level Risk Management	11
7.3	Regulatory Applications	11
8	Model Limitations and Extensions	11
8.1	Current Limitations	11
8.2	Potential Extensions	11
8.2.1	Conditional Liquidity	11
8.2.2	Optimal Execution	12
8.2.3	Network Effects	12
9	Conclusion	12
10	References	12

1 Introduction

Market liquidity represents the ease with which an asset can be bought or sold without causing significant price movement. For portfolio managers and risk managers, understanding the liquidity profile of a trading book is crucial for:

- **Risk Management:** Illiquid positions increase exit costs during market stress
- **Capital Allocation:** Liquidity costs affect expected returns
- **Regulatory Compliance:** Basel III and other frameworks require liquidity assessment
- **Portfolio Optimization:** Liquidity constraints affect optimal portfolio construction

This framework provides a systematic approach to quantifying liquidity across multiple dimensions.

2 Mathematical Framework

2.1 Position Representation

Let a trading position i be characterized by the tuple:

$$P_i = (S_i, Q_i, M_i, B_i, A_i, V_i) \quad (1)$$

where:

S_i : Security symbol/identifier

Q_i : Quantity held (positive for long, negative for short)

M_i : Current market price (mid-price)

B_i : Best bid price

A_i : Best ask price

V_i : Average daily trading volume

The position value is defined as:

$$PV_i = |Q_i| \times M_i \quad (2)$$

2.2 Bid-Ask Spread Measures

The bid-ask spread is the most fundamental liquidity measure, representing the immediate cost of a round-trip transaction.

2.2.1 Absolute Spread

$$\text{Spread}_{\text{abs},i} = A_i - B_i \quad (3)$$

2.2.2 Relative Spread

The relative spread normalizes the absolute spread by the mid-price:

$$\text{Spread}_{\text{rel},i} = \frac{A_i - B_i}{\frac{A_i + B_i}{2}} = \frac{A_i - B_i}{M_i} \quad (4)$$

This is typically expressed as a percentage.

2.2.3 Spread in Basis Points

$$\text{Spread}_{\text{bps},i} = \text{Spread}_{\text{rel},i} \times 10,000 \quad (5)$$

2.3 Amihud Illiquidity Ratio

The Amihud (2002) illiquidity measure captures the price impact per unit of trading volume:

$$\text{ILLIQ}_i = \frac{|R_i|}{V_i \times M_i} \quad (6)$$

where R_i is the daily return for security i . For position analysis without time-series data, we can approximate:

$$\text{ILLIQ}_i \approx \frac{\sigma_i}{V_i \times M_i} \quad (7)$$

where σ_i represents expected daily volatility or a standardized return assumption.

Interpretation: Higher values indicate greater illiquidity. A value of $\text{ILLIQ}_i = 10^{-6}$ means that \$1 million in trading volume generates a 1% price change.

2.4 Volume Liquidity Ratio

This metric measures how long it would take to liquidate a position relative to normal trading volumes:

$$\text{VLR}_i = \frac{|Q_i|}{V_i} \quad (8)$$

Interpretation: A value of $\text{VLR}_i = 2.5$ indicates the position size equals 2.5 days of average trading volume.

2.4.1 Time to Liquidate

Given a maximum participation rate α (fraction of daily volume):

$$T_{\text{liquidate},i} = \frac{\text{VLR}_i}{\alpha} \quad (9)$$

Standard practice uses $\alpha = 0.10$ (10% of daily volume) to avoid excessive market impact.

2.5 Market Impact Model

Market impact represents the price movement caused by executing a trade. We decompose impact into temporary and permanent components.

2.5.1 Temporary Impact

Temporary impact reflects the bid-ask spread crossing cost:

$$I_{\text{temp},i} = \frac{\text{Spread}_{\text{rel},i}}{2} \quad (10)$$

This represents the cost of immediately executing at market prices.

2.5.2 Permanent Impact

Permanent impact reflects lasting price movement from order execution. Using a simplified square-root model:

$$I_{\text{perm},i} = \beta \times \sqrt{\frac{|Q_i|}{V_i}} \quad (11)$$

where β is a market impact coefficient. A simplified linear approximation:

$$I_{\text{perm},i} \approx \gamma \times \frac{\text{VLR}_i}{\alpha} \quad (12)$$

where γ represents the price impact per day of volume (typically 0.1% - 0.5%).

2.5.3 Total Market Impact

$$I_{\text{total},i} = I_{\text{temp},i} + I_{\text{perm},i} \quad (13)$$

The monetary cost of liquidation:

$$\text{Cost}_i = I_{\text{total},i} \times \text{PV}_i \quad (14)$$

2.6 Composite Liquidity Score

We construct a normalized liquidity score $L_i \in [0, 100]$ where higher values indicate greater liquidity:

$$L_i = w_1 \times L_{\text{spread},i} + w_2 \times L_{\text{volume},i} \quad (15)$$

with weights $w_1 = 0.6$ and $w_2 = 0.4$.

2.6.1 Spread-Based Score

$$L_{\text{spread},i} = \max(0, 100 - \text{Spread}_{\text{bps},i}) \quad (16)$$

2.6.2 Volume-Based Score

$$L_{\text{volume},i} = \max(0, 100 - 20 \times \text{VLR}_i) \quad (17)$$

The coefficient 20 penalizes positions requiring more than 5 days to liquidate.

2.6.3 Liquidity Rating Classification

$$\text{Rating}(L_i) = \begin{cases} \text{Highly Liquid} & \text{if } L_i \geq 80 \\ \text{Liquid} & \text{if } 60 \leq L_i < 80 \\ \text{Moderately Liquid} & \text{if } 40 \leq L_i < 60 \\ \text{Illiquid} & \text{if } 20 \leq L_i < 40 \\ \text{Highly Illiquid} & \text{if } L_i < 20 \end{cases} \quad (18)$$

3 Portfolio-Level Aggregation

For a portfolio $\mathcal{P}$ containing N positions, we compute value-weighted metrics.

3.1 Total Portfolio Value

$$PV_{\mathcal{P}} = \sum_{i=1}^N PV_i \quad (19)$$

3.2 Weighted Average Spread

$$\overline{\text{Spread}}_{\mathcal{P}} = \frac{\sum_{i=1}^N \text{Spread}_{\text{bps},i} \times PV_i}{PV_{\mathcal{P}}} \quad (20)$$

3.3 Total Liquidation Cost

$$\text{Cost}_{\mathcal{P}} = \sum_{i=1}^N \text{Cost}_i \quad (21)$$

3.4 Portfolio Liquidity Cost Ratio

$$\text{LCR}_{\mathcal{P}} = \frac{\text{Cost}_{\mathcal{P}}}{PV_{\mathcal{P}}} \times 100\% \quad (22)$$

This represents the total expected cost to liquidate the entire portfolio as a percentage of portfolio value.

3.5 Average Liquidity Score

$$\overline{L}_{\mathcal{P}} = \frac{1}{N} \sum_{i=1}^N L_i \quad (23)$$

4 Algorithm and Flow Charts

4.1 High-Level Analysis Flow

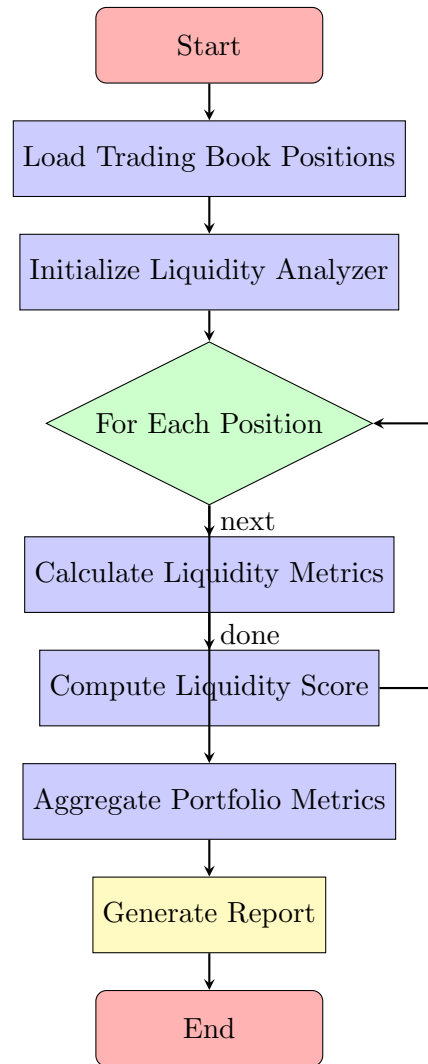


Figure 1: High-Level Liquidity Analysis Flow

4.2 Position-Level Metric Calculation

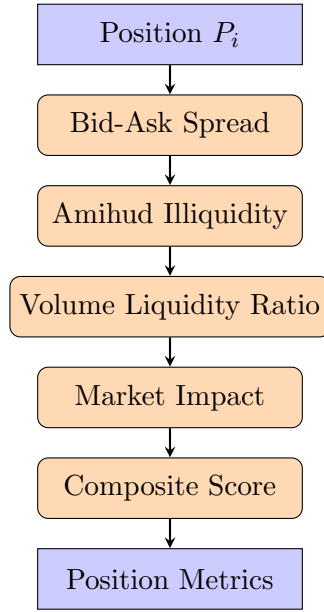


Figure 2: Position-Level Calculation Sequence

4.3 Market Impact Calculation Flow

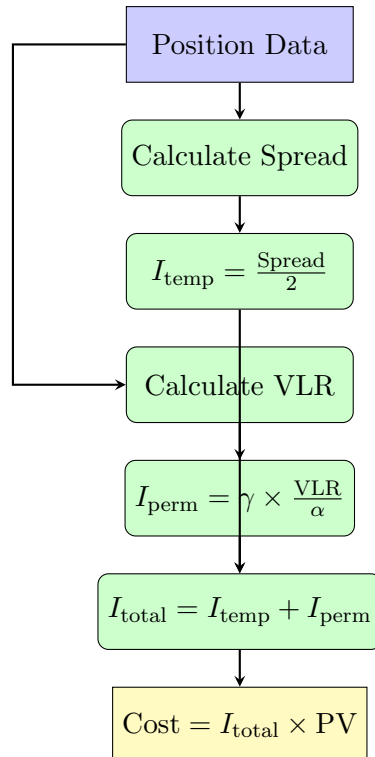


Figure 3: Market Impact Calculation Flow

4.4 Composite Score Computation

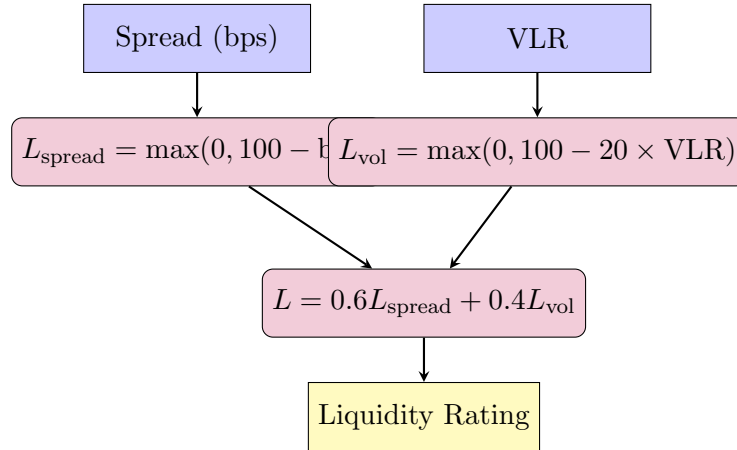


Figure 4: Composite Liquidity Score Calculation

5 Implementation Algorithm

6 Practical Example

Consider a trading book with four positions:

Table 1: Example Trading Book

Symbol	Quantity	Price	Bid	Ask	Daily Vol	Position Value
AAPL	10,000	180.50	180.48	180.52	50,000,000	\$1,805,000
TSLA	5,000	242.80	242.70	242.90	100,000,000	\$1,214,000
SMALL	50,000	15.25	15.10	15.40	100,000	\$762,500
MICRO	100,000	2.50	2.40	2.60	50,000	\$250,000

6.1 AAPL Analysis

Spread Metrics:

$$\text{Spread}_{\text{abs}} = 180.52 - 180.48 = 0.04$$

$$\text{Spread}_{\text{rel}} = \frac{0.04}{180.50} = 0.0221\%$$

$$\text{Spread}_{\text{bps}} = 2.21$$

Volume Ratio:

$$\text{VLR} = \frac{10,000}{50,000,000} = 0.0002$$

$$T_{\text{liq}} = \frac{0.0002}{0.10} = 0.002 \text{ days} < 1 \text{ hour}$$

Algorithm 1 Trading Book Liquidity Analysis

Require: Trading book $\mathcal{P} = \{P_1, P_2, \dots, P_N\}$

Ensure: Liquidity metrics for each position and portfolio aggregates

```
1: Initialize empty results list  $\mathcal{R}$ 
2: for each position  $P_i$  in  $\mathcal{P}$  do
3:    $M_i \leftarrow (B_i + A_i)/2$  ▷ Mid-price
4:    $PV_i \leftarrow |Q_i| \times M_i$  ▷ Position value
5:   Calculate spread metrics:
6:    $\text{Spread}_{\text{abs}} \leftarrow A_i - B_i$ 
7:    $\text{Spread}_{\text{rel}} \leftarrow \text{Spread}_{\text{abs}}/M_i$ 
8:    $\text{Spread}_{\text{bps}} \leftarrow \text{Spread}_{\text{rel}} \times 10000$ 
9:   Calculate Amihud illiquidity:
10:   $\text{ILLIQ}_i \leftarrow |R_i|/(V_i \times M_i)$ 
11:  Calculate volume ratio:
12:   $\text{VLR}_i \leftarrow |Q_i|/V_i$ 
13:   $T_{\text{liq}} \leftarrow \text{VLR}_i/\alpha$  ▷ Time to liquidate
14:  Calculate market impact:
15:   $I_{\text{temp}} \leftarrow \text{Spread}_{\text{rel}}/2$ 
16:   $I_{\text{perm}} \leftarrow \gamma \times T_{\text{liq}}$ 
17:   $I_{\text{total}} \leftarrow I_{\text{temp}} + I_{\text{perm}}$ 
18:   $\text{Cost}_i \leftarrow I_{\text{total}} \times PV_i$ 
19:  Calculate liquidity score:
20:   $L_{\text{spread}} \leftarrow \max(0, 100 - \text{Spread}_{\text{bps}})$ 
21:   $L_{\text{vol}} \leftarrow \max(0, 100 - 20 \times \text{VLR}_i)$ 
22:   $L_i \leftarrow 0.6 \times L_{\text{spread}} + 0.4 \times L_{\text{vol}}$ 
23:   $\text{Rating}_i \leftarrow \text{ClassifyLiquidity}(L_i)$ 
24:  Add all metrics to  $\mathcal{R}$ 
25: end for
26: Calculate portfolio aggregates:
27:  $PV_{\mathcal{P}} \leftarrow \sum_{i=1}^N PV_i$ 
28:  $\overline{\text{Spread}} \leftarrow \sum_{i=1}^N (\text{Spread}_{\text{bps},i} \times PV_i)/PV_{\mathcal{P}}$ 
29:  $\text{Cost}_{\mathcal{P}} \leftarrow \sum_{i=1}^N \text{Cost}_i$ 
30:  $\text{LCR}_{\mathcal{P}} \leftarrow 100 \times \text{Cost}_{\mathcal{P}}/PV_{\mathcal{P}}$ 
    return  $\mathcal{R}$ , portfolio aggregates
```

Market Impact:

$$I_{\text{temp}} = \frac{0.0221\%}{2} = 0.011\%$$

$$I_{\text{perm}} = 0.1\% \times 0.002 = 0.0002\%$$

$$I_{\text{total}} = 0.0112\%$$

$$\text{Cost} = 0.0112\% \times \$1,805,000 = \$202$$

Liquidity Score:

$$L_{\text{spread}} = \max(0, 100 - 2.21) = 97.79$$

$$L_{\text{vol}} = \max(0, 100 - 20 \times 0.0002) = 99.996$$

$$L = 0.6 \times 97.79 + 0.4 \times 99.996 = 98.67$$

$$\text{Rating} = \text{Highly Liquid}$$

6.2 MICRO Analysis (Illiquid Position)**Spread Metrics:**

$$\text{Spread}_{\text{abs}} = 2.60 - 2.40 = 0.20$$

$$\text{Spread}_{\text{rel}} = \frac{0.20}{2.50} = 8.0\%$$

$$\text{Spread}_{\text{bps}} = 800$$

Volume Ratio:

$$\text{VLR} = \frac{100,000}{50,000} = 2.0$$

$$T_{\text{liq}} = \frac{2.0}{0.10} = 20 \text{ days}$$

Market Impact:

$$I_{\text{temp}} = \frac{8.0\%}{2} = 4.0\%$$

$$I_{\text{perm}} = 0.1\% \times 20 = 2.0\%$$

$$I_{\text{total}} = 6.0\%$$

$$\text{Cost} = 6.0\% \times \$250,000 = \$15,000$$

Liquidity Score:

$$L_{\text{spread}} = \max(0, 100 - 800) = 0$$

$$L_{\text{vol}} = \max(0, 100 - 20 \times 2.0) = 60$$

$$L = 0.6 \times 0 + 0.4 \times 60 = 24$$

$$\text{Rating} = \text{Illiquid}$$

7 Interpretation and Usage**7.1 Position-Level Decisions**

- $L \geq 80$: Can be liquidated quickly with minimal market impact. Suitable for high-turnover strategies.

- $60 \leq L < 80$: Normal liquidity. Standard execution algorithms appropriate.
- $40 \leq L < 60$: Requires careful execution over multiple days. Consider VWAP/TWAP algorithms.
- $20 \leq L < 40$: Significant liquidity constraints. May require weeks to exit. Consider position limits.
- $L < 20$: Highly illiquid. Requires specialized broker relationships or block trading.

7.2 Portfolio-Level Risk Management

The portfolio liquidity cost ratio (LCR) provides a single metric for overall book liquidity:

$$\text{Risk}(\text{LCR}_{\mathcal{P}}) = \begin{cases} \text{Low} & \text{if } \text{LCR} < 0.5\% \\ \text{Moderate} & \text{if } 0.5\% \leq \text{LCR} < 2.0\% \\ \text{High} & \text{if } 2.0\% \leq \text{LCR} < 5.0\% \\ \text{Extreme} & \text{if } \text{LCR} \geq 5.0\% \end{cases} \quad (24)$$

7.3 Regulatory Applications

Basel III Liquidity Coverage Ratio (LCR): Positions with high liquidation costs and long liquidation times should receive appropriate haircuts when calculating high-quality liquid assets (HQLA).

SEC Rule 22e-4 (Mutual Fund Liquidity): US mutual funds must classify positions into liquidity buckets. This framework maps directly:

- Highly Liquid: $T_{\text{liq}} \leq 3$ days
- Moderately Liquid: $3 < T_{\text{liq}} \leq 7$ days
- Less Liquid: $7 < T_{\text{liq}} \leq 15$ days
- Illiquid: $T_{\text{liq}} > 15$ days

8 Model Limitations and Extensions

8.1 Current Limitations

1. **Static Analysis:** Does not account for time-varying liquidity (intraday patterns, market stress)
2. **Simplified Impact Model:** Real market impact exhibits non-linear behavior
3. **No Correlation:** Ignores correlation between positions during fire-sale scenarios
4. **Single Market:** Does not account for multi-venue or international execution

8.2 Potential Extensions

8.2.1 Conditional Liquidity

Incorporate regime-switching models:

$$\text{Spread}_i(t) = \text{Spread}_{\text{normal},i} \times (1 + \lambda_t) \quad (25)$$

where λ_t represents a market stress factor.

8.2.2 Optimal Execution

Integrate with Almgren-Chriss optimal execution framework:

$$\text{Cost}_{\text{opt}} = \frac{\sigma^2 \tau}{2\eta} + \frac{\eta Q^2}{2\tau} \quad (26)$$

where τ is execution time, η is temporary impact parameter, and σ is volatility.

8.2.3 Network Effects

Model portfolio-wide liquidation using fire-sale externalities:

$$\text{Impact}_{\text{portfolio}} > \sum_{i=1}^N \text{Impact}_i \quad (27)$$

9 Conclusion

This framework provides a comprehensive, mathematically rigorous approach to measuring trading book liquidity. By combining multiple established metrics into a unified system, it enables quantitative risk management, portfolio optimization under liquidity constraints, and regulatory compliance.

The composite liquidity score offers a simple summary metric while preserving the detailed multi-dimensional analysis needed for sophisticated applications. Implementation in Python enables rapid analysis of large trading books and integration into existing risk management systems.

10 References

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